

Week 7 HW Question 2:

Date:

Given: $u = (-3, \frac{1}{2})$, $v = (1, -\frac{1}{8})$

$$a.) \|u\| = \sqrt{(-3)^2 + (\frac{1}{2})^2} = \sqrt{9 + \frac{1}{4}} = \sqrt{\frac{37}{4}} = \frac{\sqrt{37}}{2}$$

$$b.) \|v\| = \sqrt{(1)^2 + (-\frac{1}{8})^2} = \sqrt{1 + \frac{1}{64}} = \sqrt{\frac{65}{64}} = \frac{\sqrt{65}}{8}$$

$$c.) u + v = (-3 + 1, \frac{1}{2} - \frac{1}{8}) = (-2, \frac{3}{8})$$

$$\|u + v\| = \sqrt{(-2)^2 + (\frac{3}{8})^2} = \sqrt{4 + \frac{9}{64}} = \sqrt{\frac{265}{64}} = \frac{\sqrt{265}}{8}$$